

Estimation of Cluster-Specific Causal Effects on Spatially Associated Survival Data Using SoftBART

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1. Big Picture & Motivation

Breast cancer: Most frequently diagnosed cancer; 2nd leading cause of cancer death among US women.

Treatment Delay (TD): Interval between diagnosis and first treatment. Delays >90 days significantly worsen 5-year survival rates.

Florida Cancer Registry (FCR): $N=68,009$ patients $\times K=67$ counties; 41.3% censoring; 6 strata (2 races $\times$ 3 stages).

Goal: Estimate county-specific and patient-specific **causal effects** of short TD (≤ 90 days) on survival while accommodating spatial association.

2. Key Challenges

- Spatial dependence** among neighboring counties
- Non-linear** treatment–confounder–outcome relationships
- Right-censored** survival data (41.3% censoring)
- Non-random treatment** assignment with confounding
- Clustered structure:** patients nested within counties
- Existing methods ignore spatial association or require rigid parametric forms

3. Proposed Method: PS-LND (SBART + DAGAR)

A **two-stage Bayesian** approach:

Stage 1: Propensity Score

$$P(Z_{ij}=1 \mid \mathbf{x}_{ij}, V_i) = \Phi(b(\mathbf{x}_{ij}) + V_i)$$

$b(\cdot)$: **SBART** prior; V : **DAGAR** spatial prior
Output: estimated PS $\hat{e}_{ij}$

Stage 2: Outcome Model (Log-Normal AFT)

$$\log T_{ij} \sim N(\underbrace{\mu(\mathbf{x}_{ij}, \hat{e}_{ij})}_{\text{prognostic}} + \underbrace{\tau(\mathbf{x}_{ij})}_{\text{treatment}} Z_{ij} + \underbrace{W_i}_{\text{spatial}}, \sigma^2)$$

μ, τ : **SBART** priors (BCF structure); W : **DAGAR** prior

Key Estimand — CERM:

$$\text{CERM}_{ij}(t^*) = \mathbb{E}[\{\{T_{ij}(1) \wedge t^*\} - \{T_{ij}(0) \wedge t^*\} \mid \mathbf{x}_{ij}, \text{Cluster} = i\}]$$

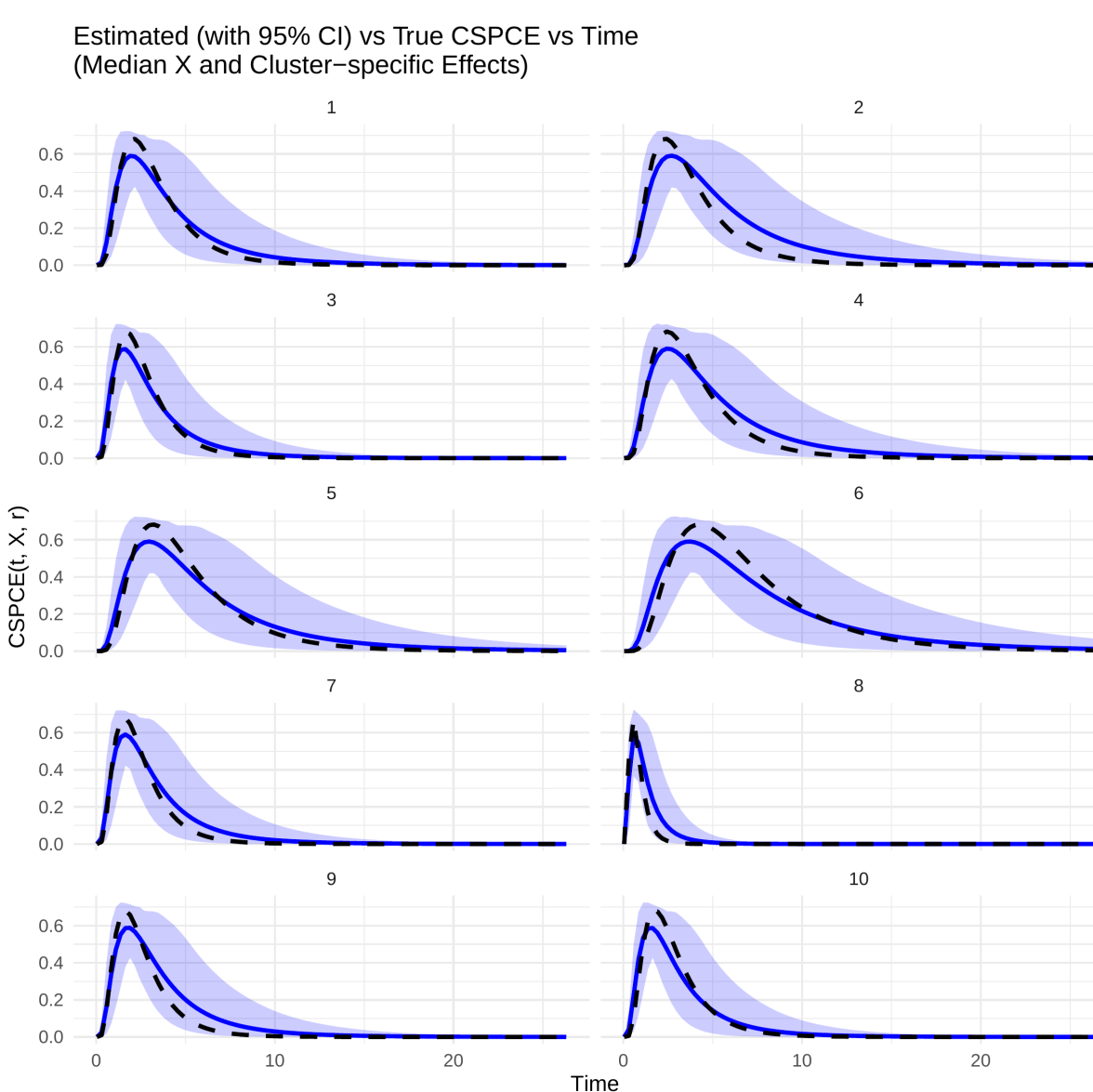
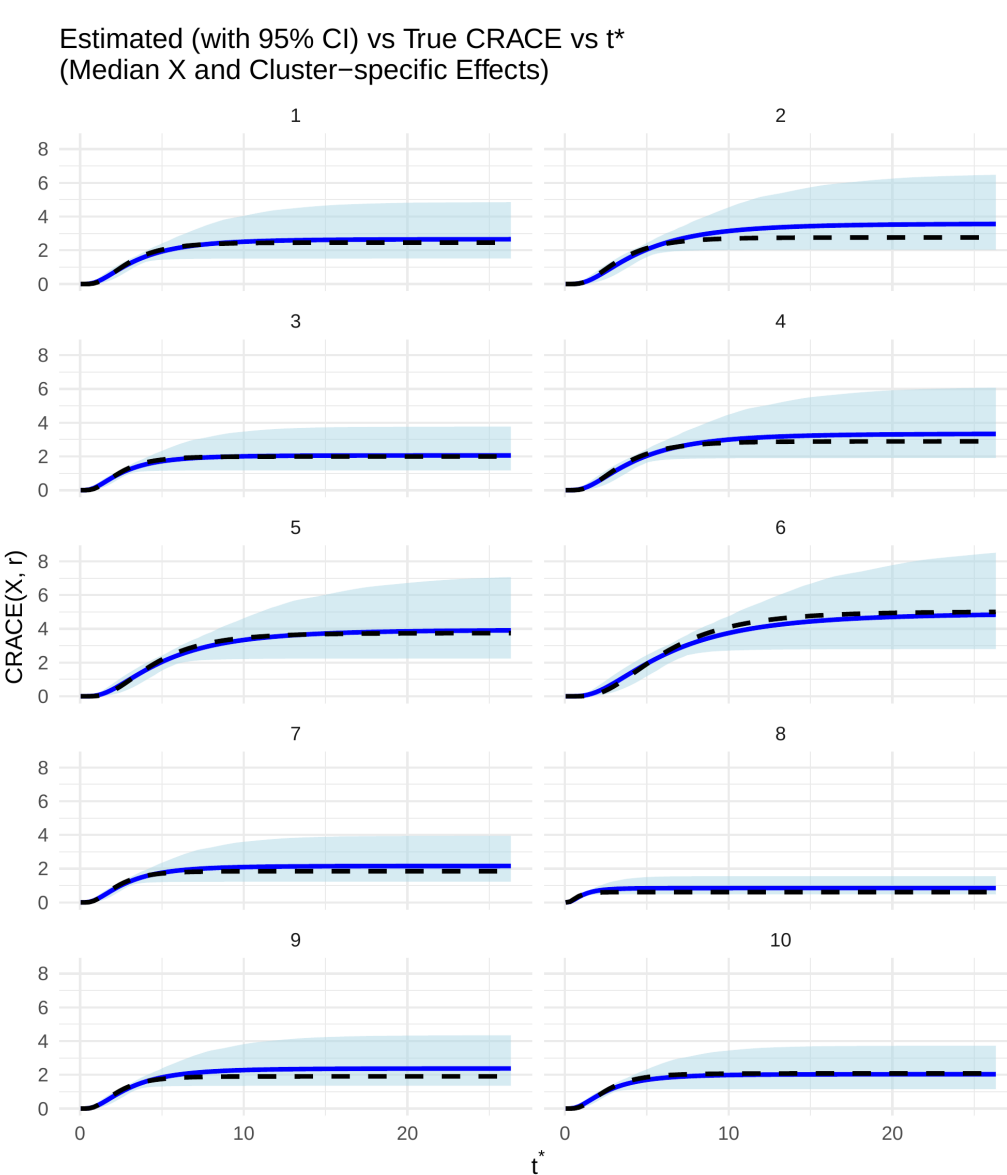
Advantages: (i) No feedback issue; (ii) SBART captures smooth nonlinearities; (iii) DAGAR guarantees proper distribution with interpretable $\rho \in (0, 1)$.

4. Simulation Study

$K=35$ clusters, $N=1,750$, 5 confounders, 150 replications.

Scenario	Metric	PS-LND	riAFTBART	PSW-Fr.	IPW
1: Correctly specified	Cov. (CI.4, $t=15$)	0.93	0.46	0.97	0.39
	MAE (CI.4, $t=15$)	0.01	0.58	0.23	0.07
2: Mis-specified	Cov. (CI.10, $t=5$)	0.95	0.89	0.64	0.53
	MAE (CI.10, $t=15$)	0.02	0.56	0.49	0.07
3: Weibull + 50% cens.	Cov. (CI.4, $t=15$)	0.90	0.67	0.76	0.42
	MAE (CI.4, $t=5$)	0.03	0.37	0.10	0.58

PS-LND: best coverage & lowest MAE across all scenarios.



5. FCR Results: Patient-Level Causal Effects (Broward County)

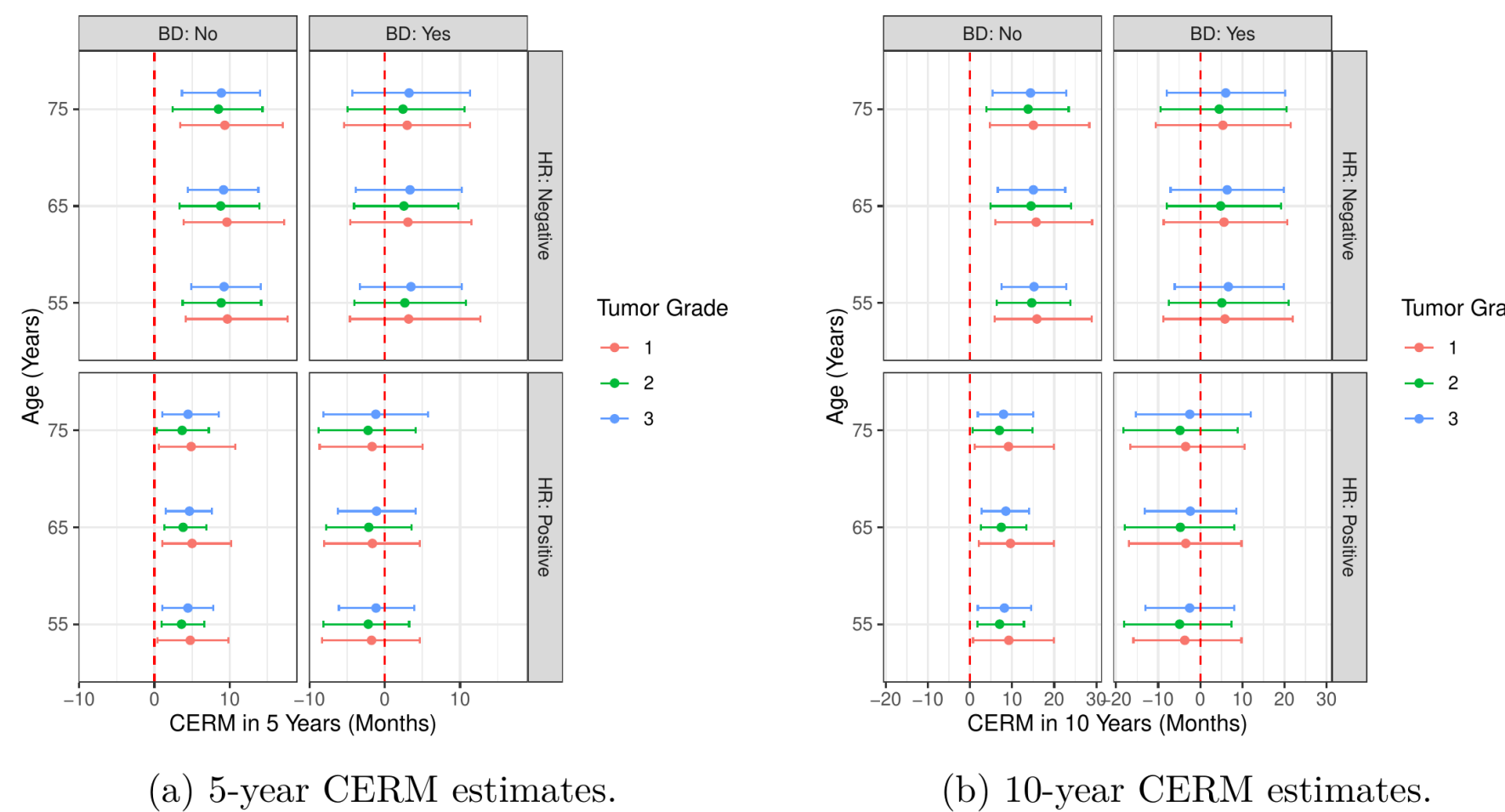


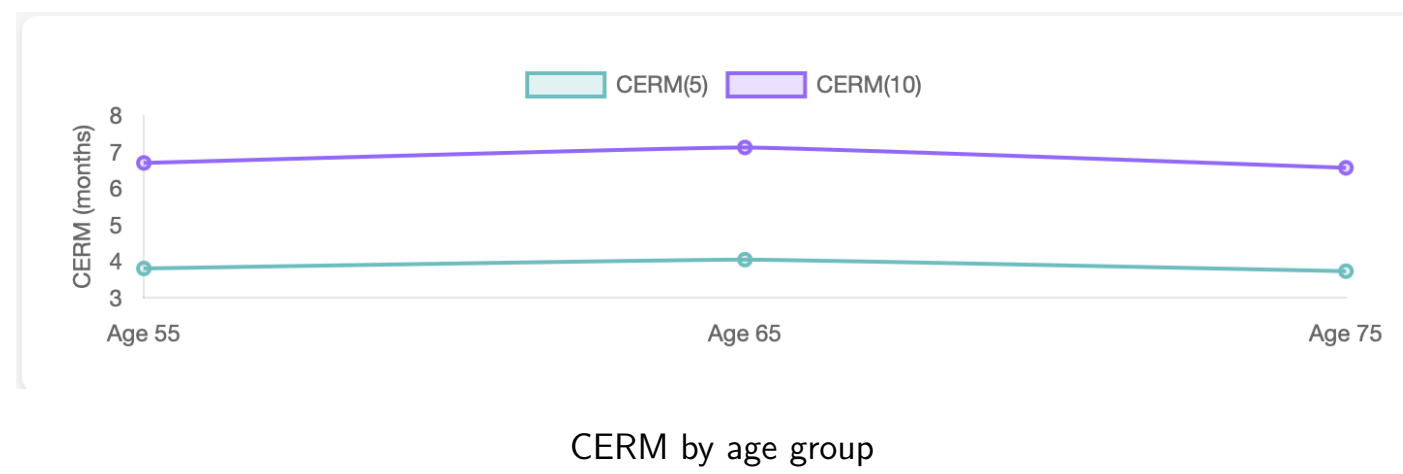
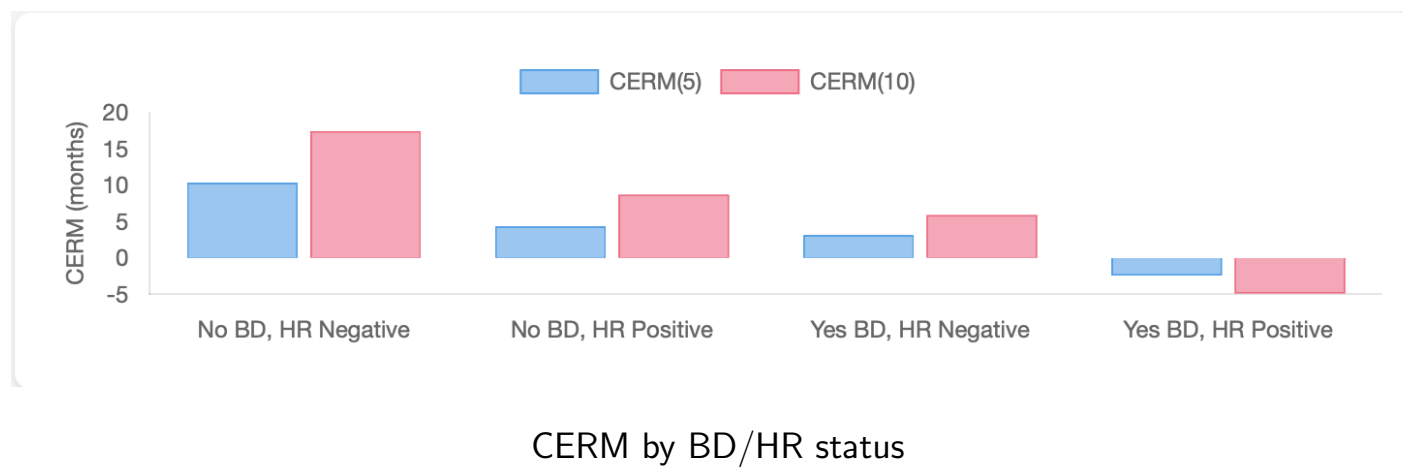
Fig 1: Point and 95% interval estimates of CERM (in months) restricted within intervals of (a) 5 and (b) 10 years for WA patients with Distant-stage cancer in Broward County

Point & 95% CI of CERM (months) at 5 and 10 years for WA-Distant patients ($n=183$) across 36 covariate combinations.

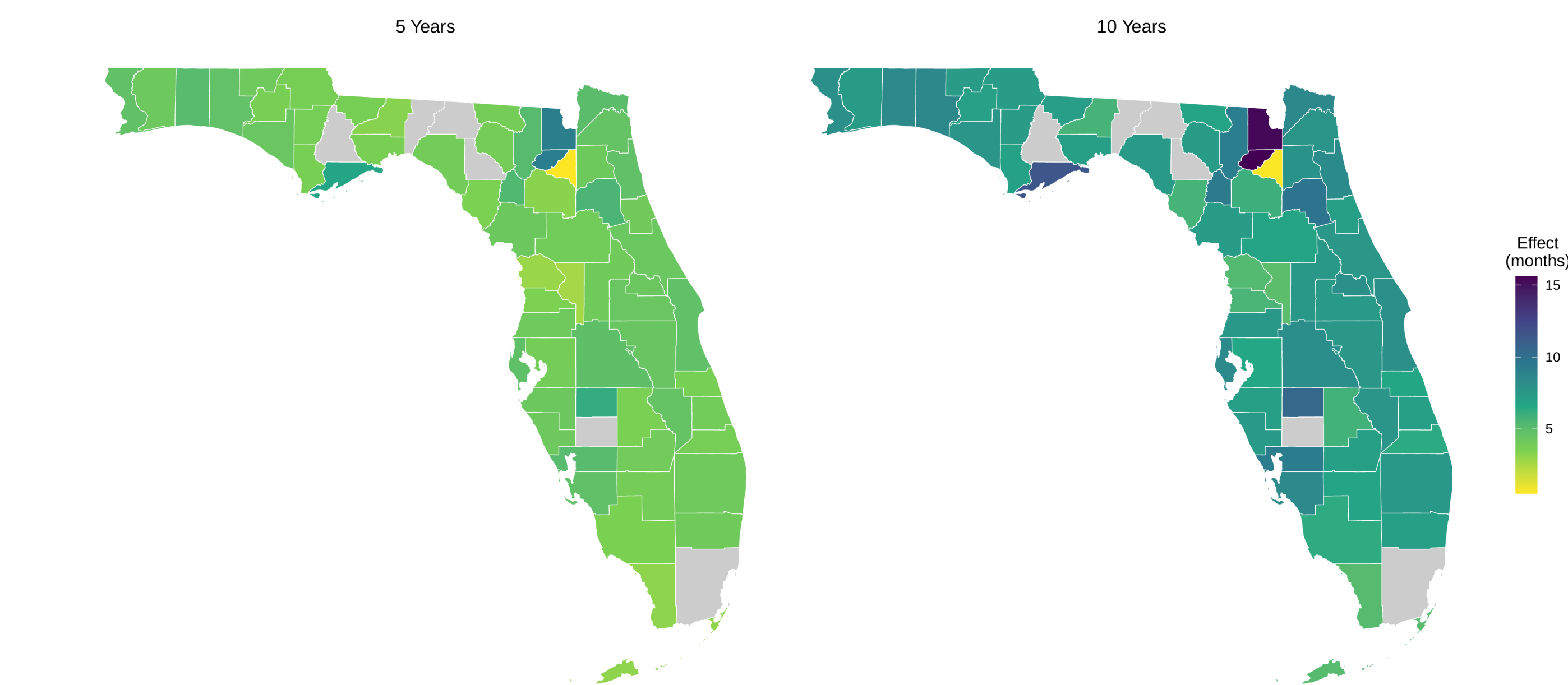
HR–, no BD: strongest benefit (**13.8–15.9 mo.** at 10 yr)

HR+, no BD: moderate (**6.9–9.6 mo.**)

With BD: **no significant effect** (CI $\ni 0$)



6. FCR Results: County-Level Spatial Patterns

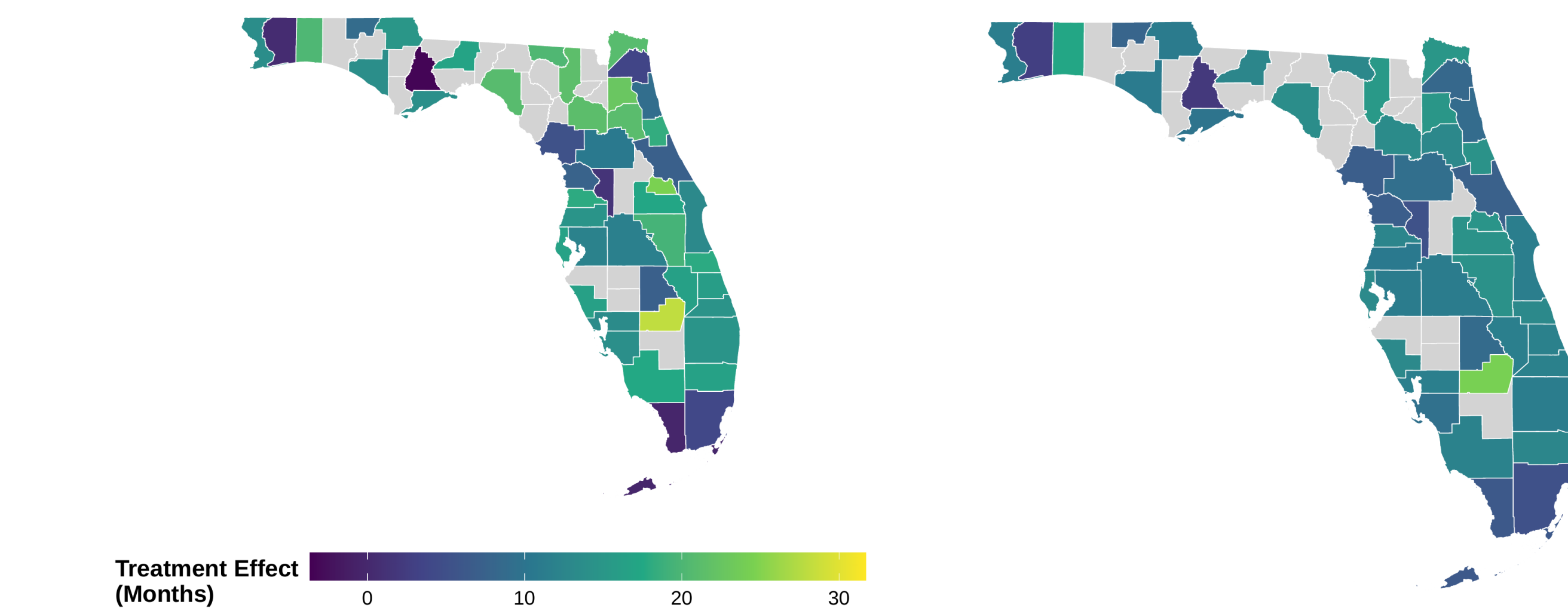


ACERM, at 5 (left) and 10 (right) years. Yellow ≈ 0 mo. $\rightarrow$ green = 15+ mo.; gray = no data.

Highest: Union (>15 mo.), Baker (>13 mo.)

Near-zero: Bradford (CI $\ni 0$)

Clear **spatial clustering** of treatment effects



ACERMT, for treated patients. Glades County >14 months.

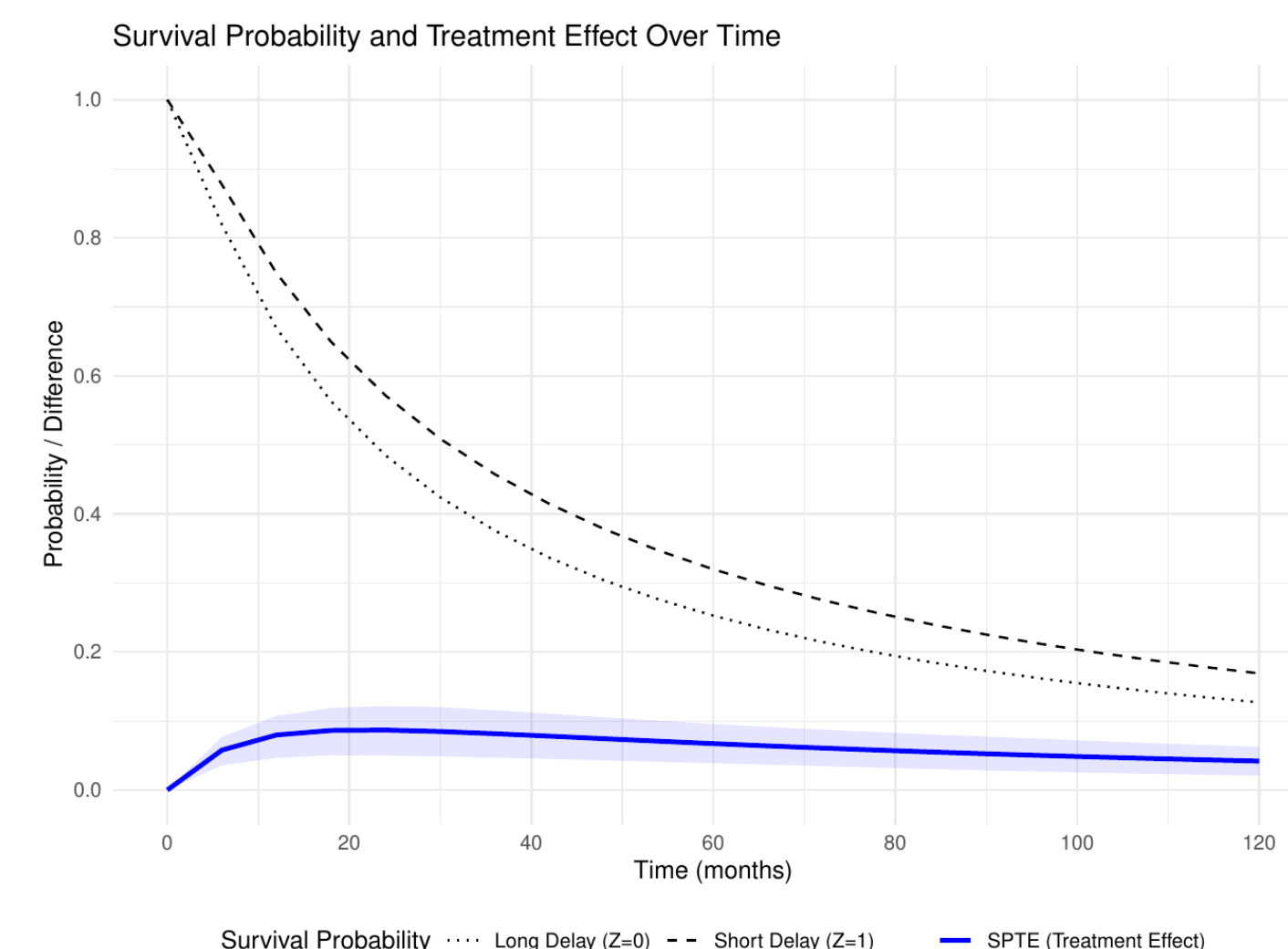
7. FCR Results: State-Wide Treatment Effects

Estimand	Interval	N	Est. (mo.)	95% CI
RATE	10 yr	1994	7.62	(4.42, 10.86)
RATT	10 yr	687	6.74	(3.09, 10.08)
RATU	10 yr	1307	8.09	(4.60, 11.42)
RATE	5 yr	1994	4.43	(2.56, 6.23)
RATT	5 yr	687	3.93	(2.15, 5.64)
RATU	5 yr	1307	4.70	(2.75, 6.51)

No CI contains zero $\Rightarrow$ strong evidence of survival benefit

RATU > RATE > RATT: patients with **long TD stand to gain the most**

5-yr effects are $\approx 58\%$ of 10-yr effects $\Rightarrow$ benefits **persist beyond 5 years**



$\widehat{\text{SPTE}}(t)$: Causal effect on survival probability over time. Peaks at **11%** at $t=2.2$ years; entire 95% CI above 0 throughout follow-up.

8. Conclusions & Policy Implications

- PS-LND** outperforms riAFT-BART, PSW-Frailty, and IPW in all simulations
- Short TD (≤ 90 days) provides significant survival benefit: **7.6 months** average gain over 10 years for WA-Distant patients in FL
- Significant spatial heterogeneity:** effects vary from near-zero to >15 months across counties
- HR– patients without biopsy delay** benefit most — priority targets for policy intervention
- Patients currently experiencing **long TD have the most to gain** from intervention (RATU > RATT)
- Results support **localized Patient Navigation Programs** to reduce TD in high-impact counties

9. References & Acknowledgments

Refs: Bhattacharya et al. (2025), *AOAS* • Chipman et al. (2010) • Datta et al. (2019) • Hahn et al. (2020) • Hu & Lin (2022) **Code:** github.com/indrabati646/PS-LND-DAGAR **Funding:** NIH R03CA205018-01, R21DE034124-01A1; NSF 2413549